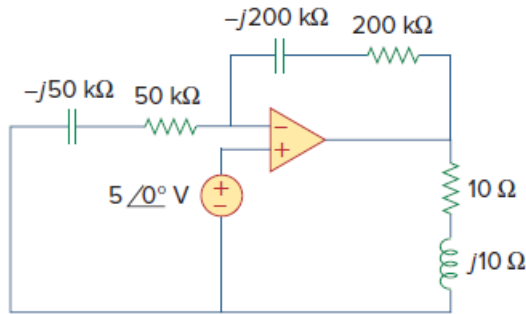


Problem

Obtain the average power absorbed by the $10\text{-}\Omega$ resistor in the op amp circuit in Fig. 11.85.

Figure 11.85:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.85 in the text book.

The voltage at the inverting terminal is equal to the $5\angle 0^\circ \text{ V}$ due to the virtual voltage available at the non-inverting terminal. Assume the voltage at the output terminal of the operational amplifier to be v_o .

Apply Kirchhoff's Current Law at output terminal of the operational amplifier.

$$\begin{aligned} \frac{v_o - 5\angle 0^\circ}{200 \text{ k} - j200 \text{ k}} + \frac{v_o}{10 + j10} &= 0 \\ v_o \left(\frac{1}{200 \text{ k} - j200 \text{ k}} + \frac{1}{10 + j10} \right) - \frac{5\angle 0^\circ}{200 \text{ k} - j200 \text{ k}} &= 0 \\ v_o (2.5 \mu + j2.5 \mu + 50 \text{ m} - j50 \text{ m}) &= 12.5 \mu + j12.5 \mu \\ v_o (50 \text{ m} - j50 \text{ m}) &= 12.5 \mu + j12.5 \mu \\ v_o &= \frac{12.5 \mu + j12.5 \mu}{50 \text{ m} - j50 \text{ m}} \\ &= j250 \mu \\ &= 250\angle 90^\circ \mu\text{V} \end{aligned}$$

Therefore, the value of the output voltage is $250\angle 90^\circ \mu\text{V}$.

Compare $250\angle 90^\circ \mu\text{V}$ with $V_m \angle \theta_v^\circ \text{ V}$.

The value of V_m and θ_v are $250 \mu\text{V}$ and 90° respectively.

[Comments \(1\)](#)



Anonymous

This is incorrect...you cannot write a KCL equation at v_o unless you take into account the output current

Step 2 of 3

Calculate the value of the current passing through $10\ \Omega$ resistor.

$$i_o = \frac{v_o}{10 + j10}$$

Substitute $250\angle 90^\circ\ \mu\text{V}$ for v_o .

$$\begin{aligned} i_o &= \frac{250\angle 90^\circ\ \mu}{10 + j10} \\ &= \frac{250\angle 90^\circ\ \mu}{14.14\angle 45^\circ} \\ &= 17.68\angle 45^\circ\ \mu\text{A} \end{aligned}$$

Therefore, the value of the current passing through $10\ \Omega$ resistor is $17.68\angle 45^\circ\ \mu\text{A}$.

Compare $17.68\angle 45^\circ\ \mu\text{A}$ with $I_m\angle\theta_i\ \text{V}$.

The value of I_m and θ_i are $17.68\ \mu\text{A}$ and 45° respectively.

Step 3 of 3

Calculate the average power dissipated in the $10\ \Omega$ resistor.

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

Substitute $17.68\ \mu\text{A}$ for I_m , 45° for θ_i , $250\ \mu\text{V}$ for V_m and 90° for θ_v .

$$\begin{aligned} P &= \frac{1}{2} (250\ \mu)(17.68\ \mu) \cos(90^\circ - 45^\circ) \\ &= (2.21\ \text{n}) \cos(45^\circ) \\ &= (2.21\ \text{n})(0.707) \\ &= 1.56\ \text{nW} \end{aligned}$$

Therefore, the value of the average power absorbed by $10\ \Omega$ resistor is 1.56 nW.